

GOLDEN HOUR

Samaritan Shield

Architecture, Technical Stack and Application Flow

A detailed, source-grounded technical dossier of the supplied project: citizen SOS, hospital dispatch, control-room operations, CCTV detection, voice triage, realtime communications, data storage, and traffic simulation.

Prepared 23 September 2026

Coverage	Reference depth
3 roles / 3 application processes	HTTP and socket contracts, state ownership, integrations
9 persisted models	All fields, defaults, indexes, and nested record shapes
Implementation + verification	Actual check outputs, prototype behavior, and missing pieces

Read sections 1-13 for the system explanation. Use the appendices for exact technical declarations and source traceability. This is a technical reference, not a live incident-data export.

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1. Project identity and coverage

Golden Hour, branded Samaritan Shield, is an emergency-response prototype connecting citizen responders, hospital desks, and a city control room. Its core operation is to record an emergency, merge nearby reports, locate a hospital, guide the responder through an encoded first-aid flow, and show incident status and telemetry to the hospital. A separate Python service detects possible road accidents from camera footage and feeds the same incident service.

This report documents the supplied `Golden_hour-main` filesystem snapshot, inspected on 23 September 2026. This directory has no Git metadata, so no commit identifier or history can be established. The code, manifests, schemas, scripts, tests, assets, and configuration templates are the evidence. Source paths throughout are relative to the project root.

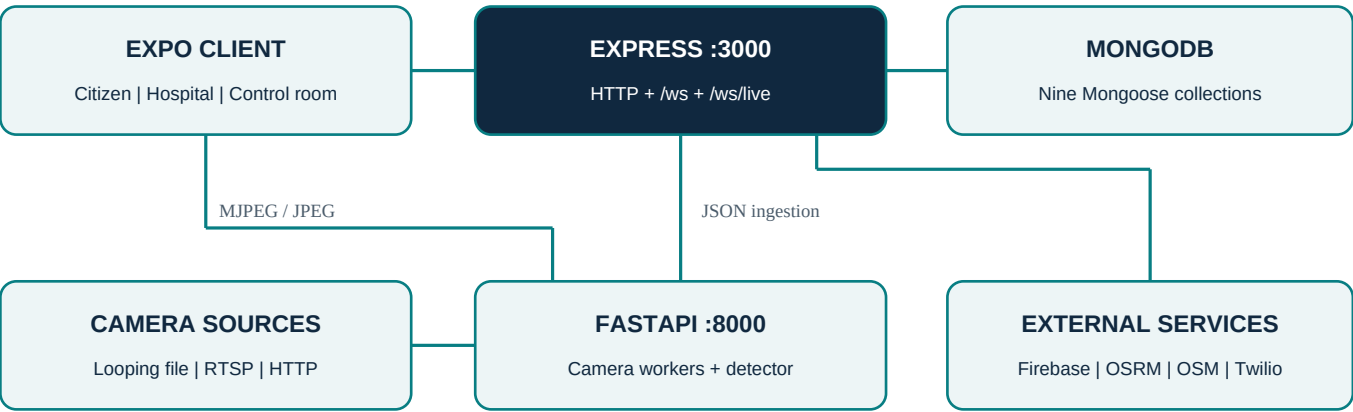
Coverage includes all three application roles, all discovered HTTP routes and WebSocket protocols, all nine database models, every declared direct dependency, all Python settings, environment-variable references, launch and provisioning paths, test coverage, bundled media, and implementation gaps. It is technical documentation, not an export of live MongoDB records or a reproduction of every source line. Private keys, actual `.env` values, runtime logs containing personal information, installed dependencies, compiled caches, and binary contents are deliberately not copied. Nested dependency resolution is covered by the lockfile inventory and fingerprints, rather than thousands of third-party source files.

Evidence labels: "Implemented" means found in the source; "verified" is reserved for a check actually executed during this report; "planned" refers to the implementation specification without matching files; "limitation" identifies a concrete constraint in this checkout. Medical and legal language in the application is described as application content, not independently validated guidance.

Sources: `README.md`; `app/App.tsx`; `server/server.ts`; `ml/main.py`; `docs/ANTIGRAVITY_IMPLEMENTATION_SPEC.md`.

2. Architecture

The system has three application processes and one persistence service. Expo renders the role-specific client. Express owns emergency business logic, authorization, PDF generation, and two WebSocket channels. FastAPI owns camera capture and inference and exposes JPEG/MJPEG media. MongoDB stores accounts, incidents, infrastructure, detection events, and corridors. Firebase Auth is an external identity provider in configured mode; a deliberately insecure local identity mode supports demonstrations.



Solid lines show logical data connections, not a physical network topology.

Figure 1. Three application processes with a shared backend-owned database.

Boundaries and entry points

Client: `app/package.json` points to `expo/AppEntry.js`, which loads `app/App.tsx`. React component state selects the authentication screen or the citizen, hospital, or control-room experience. There is no React Navigation dependency or separate routed frontend application. UI, interaction state, and extensive styling are colocated in the main screens.

Backend: `server/server.ts` builds Express, installs CORS and JSON parsing, mounts `/api/ml`, `/api/control`, and `/api/traffic`, declares the remaining routes, initializes authentication and telephony, connects Mongoose, attaches `/ws` and `/ws/live`, then listens on port 3000 by default. Startup occurs only when the file is executed as the main module, allowing tests to import the application.

ML: `uvicorn imports ml/main.py:app`. FastAPI lifespan requires `ML_SERVICE_TOKEN`, starts NodeBridge, fetches enabled cameras from Express, and starts a worker thread per camera. Shutdown requests worker stops and closes the bridge client. A roster change requires worker/service restart; there is no camera-management write API in this checkout.

Database: MongoDB is accessed by Express/Mongoose, not directly by the client or Python detector. GeoJSON coordinates are stored as `[longitude, latitude]`; API/UI coordinate objects generally use `{lat, lng}`. Mixing these representations is a significant integration risk.

State ownership and concurrency

MongoDB is authoritative for persisted incident status, reporter membership, certificates, roles, hospital capacity, cameras, signals, detections, and corridors. React owns current screen, in-progress SOS, current incident ID, PDF bytes, triage counters, and call UI. The citizen profile is stored in browser `localStorage` under `samaritan_civilian_profile`; native state has no equivalent persistent store here. Local development identity is a module variable in `app/session.ts`.

Express uses asynchronous request handlers and fire-and-forget emergency notification work. WebSocket connections and signaling user registries live in process memory. ML candidate and escalation cooldown maps also live in memory. Corridor progression uses a one-second interval. These structures are not shared across multiple Node instances and do not survive a restart.

Python uses one daemon camera thread per camera, plus one daemon bridge thread draining a bounded queue. Each camera has its own tracker and detection pipeline. A lock protects published JPEG bytes from simultaneous capture and stream reads. The optional classifier instance is shared by camera workers. The bridge queue holds 64 messages; overflow and HTTP failures are logged and dropped, with no durable retry.

External integrations

Firebase provides email/password and web Google authentication; firebase-admin validates ID tokens. Overpass/OpenStreetMap discovers hospitals, with a hardcoded Jaipur registry fallback. OSRM supplies road geometry and route duration, with a straight-line fallback. Google Maps links open external directions. BigDataCloud and Nominatim reverse-geocode coordinates; FreeIPAPI and ipapi.co supply approximate fallback positions. Google Fonts supplies web fonts. Google STUN servers assist browser WebRTC. Optional Twilio calls and SMS notify a hospital number; without credentials, the console provider logs the action and the UI still gets a live banner.

No ambulance fleet API, physical traffic-controller adapter, message broker, distributed scheduler, blockchain, electronic medical record integration, or deployed infrastructure definition was found.

3. Technical stack

Frontend

TypeScript, React 19.1.0, React Native 0.81.5, and Expo ~54.0.37 provide a shared web/iOS/Android codebase. React DOM 19.1.0 and react-native-web ^0.21.0 support browser rendering. Expo Location provides native positioning; Expo Speech and browser speech APIs provide spoken guidance; expo-speech-recognition is loaded optionally for native recognition. Expo Haptics supports tactile CPR feedback. react-native-svg supplies custom icons and the traffic simulation; qrcode generates URL/vCard identity codes. Firebase ^12.18.0 is the client authentication SDK.

Browser-specific APIs include geolocation, localStorage, Web Speech recognition/synthesis, Web Audio, Notifications, WebSocket, getUserMedia, RTCPeerConnection, Blob, and object URLs. Browser VoIP is implemented; native call support is not equivalent. Speech answers are unavailable in Expo Go unless a suitable native development build supplies the recognition module.

Frontend compilation uses strict TypeScript, ESNext modules, bundler resolution, react-jsx, DOM/ES2020 libraries, and noEmit. Babel uses the Expo preset. There is no Redux or other separate global-state library. theme.ts defines shared colors, fonts, spacing, radii, and shadows; icons.tsx defines the reusable SVG icon set. The Expo manifest declares a dark interface preference, while shared component tokens use predominantly light surfaces.

Backend

Node.js 22+ is the documented baseline; Express ^4.21.0 handles HTTP, Mongoose ^8.8.0 models MongoDB, ws ^8.21.3 handles sockets, pdf-lib ^1.17.1 creates incident PDFs, firebase-admin ^14.3.0 verifies identities, and Twilio ^5.10.4 supplies optional telephony. cors and dotenv handle origin policy and environment loading. crypto and https are Node built-ins.

The server uses strict TypeScript compiled to CommonJS/ES2020 in dist, with declaration files and source maps. Development uses ts-node. The start script compiles before executing dist/server.js. Tests use node:test, node:assert, Supertest, and direct HTTP/WebSocket clients. Test files are excluded from the production compiler input.

Computer vision

Python 3.11 is the project's documented runtime. FastAPI 0.115.6 and Uvicorn 0.34.0 expose service/media endpoints. OpenCV headless 4.10.0.84 decodes, annotates, and JPEG-encodes frames. Ultralytics 8.3.55 loads the bundled YOLOv8n detector and ByteTrack; lap 0.5.13 supplies assignment support. NumPy 1.26.4 supports numeric processing, httpx 0.28.1 sends backend requests, and Pydantic/Pydantic Settings load configuration.

TensorFlow 2.18.0, scikit-learn 1.6.0, and Matplotlib 3.10.0 are listed separately as optional training dependencies. The runtime loader supports a .keras accident classifier, but neither trained accident weights nor the referenced training scripts are present in this checkout. YOLOv8n object detection weights are present; they are not accident-classification weights.

The dependency appendix gives every direct package and version constraint, lockfile-resolved versions, and npm scripts. Constraints describe this repository, not a claim about current upstream releases.

4. Authentication and role flow

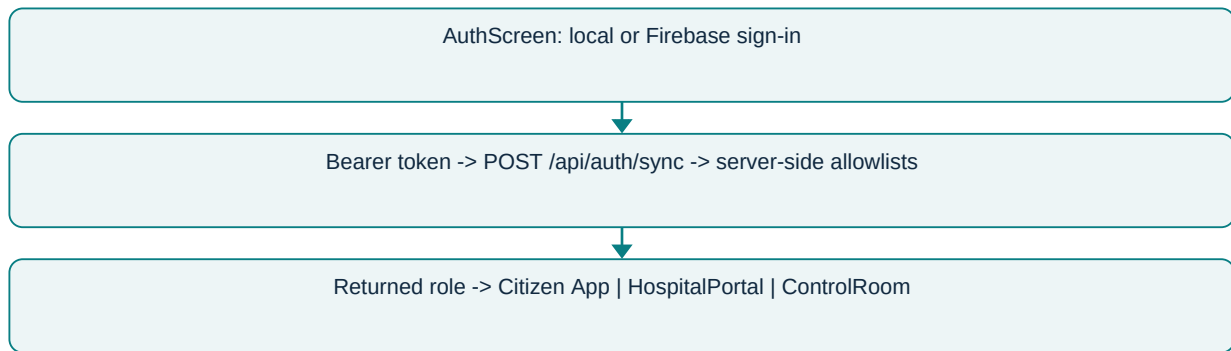


Figure 2. Role selection follows the synchronized server profile.

On launch, App renders AuthScreen until a user profile is returned. The auth screen supports email login/registration, Google login where supported, and local demo staff access. After authentication, firebaseConfig.ts calls POST /api/auth/sync using a bearer token; backend-returned role information controls which portal appears.

The frontend chooses Firebase only when all six public Firebase settings exist and local mode is not requested. Otherwise it selects local mode and displays a development banner. A local UID is deterministically derived from a normalized email with non-alphanumeric characters removed. Password checking in that mode is not authentication. Different punctuation in emails can therefore collapse into the same local UID.

In verified backend mode, requireAuth verifies Firebase ID tokens and obtains uid/email from the token. In insecure development mode only, it parses uid:email. The backend permits that mode only when explicitly enabled, not in NODE_ENV=production, and when real credentials did not initialize Firebase. Missing required authentication or ML service configuration stops startup.

Auth sync preserves an existing non-citizen role; otherwise HospitalStaff is checked by lowercase email, then ControlRoomStaff. A user cannot grant themselves a hospital or control-room role by supplying a body field. Hospital records carry hospitalId/name; control-room accounts carry a zone. Optional finite login coordinates are stored in both scalar fields and a GeoJSON point.

Hospital capacity writes are bound to the caller's hospital. Reporter read/triage APIs check that the token UID is the primary or a secondary reporter. However, the hospital incident feed and status mutation enforce the hospital role without restricting incidents to the caller's assigned hospital. Control-room REST lists are not filtered by the user's zone; selected live detection broadcasts are zone-filtered. These are the actual authorization boundaries.

Sources: app/AuthScreen.tsx; app/config.ts; app/firebaseConfig.ts; app/api.ts; app/session.ts; server/middleware/auth.ts; server/server.ts.

5. Citizen app flow

Screens and local state

The citizen has four tabs: HUB (SOS and current response), MAPS (coordinate/radar display and external hospital directions), INTEL (the severe-trauma checklist), and REPORTS (incident digest and document access). Profile editing, QR preview, certificate display, incoming-call controls, and citizen-alert banners are additional overlays. The MAPS tab is a styled placeholder/radar panel rather than a live embedded city map or real ambulance tracker.

Six categories are presented: Medical, Cardiac, Accident, Fire Force, Cops, and Airway. They change UI selection/copy; the SOS payload is only lat/lng, and the backend does not implement six distinct service-routing workflows. Search and display controls do not imply server search endpoints.

Location acquisition

The initial coordinate is a Jaipur demonstration point, 26.9090 latitude and 75.7325 longitude. The app attempts high-accuracy browser GPS (6-second timeout), standard browser GPS (5 seconds with a 30-second cached-position allowance), Expo Location with foreground permission, FreeIPAPI, and ipapi.co. Reverse geocoding adds a place label. Browser watchPosition supplies subsequent updates.

If every attempt fails, fetchLivePosition returns the existing coordinates, which may still be the demo point. Approximate IP location is not an exact incident position. The GPS watcher effect depends on a callback that depends on coordinates, so coordinate updates can cause effect teardown/recreation. This report does not treat the initial point as verified real-user location.

SOS sequence

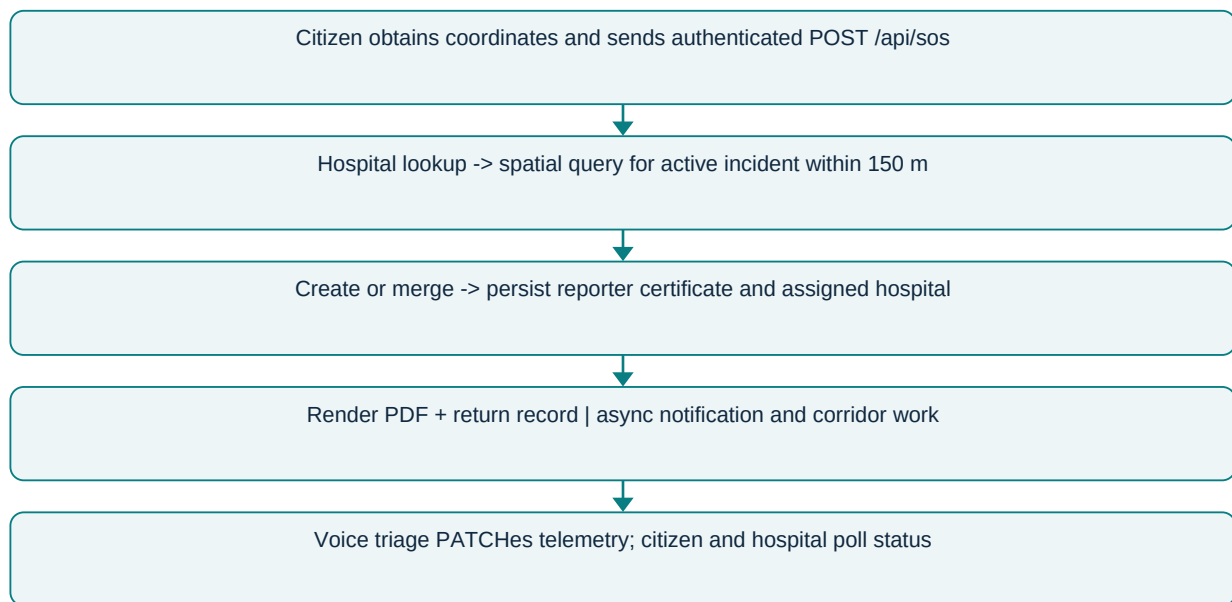


Figure 3. Citizen incident creation and ongoing synchronization.

The UI moves idle -> activating -> active. It vibrates and speaks an emergency announcement before the network request has succeeded, then obtains coordinates and POSTs /api/sos. On success it stores incident ID/code, reporter role, digest, timestamp, Base64 PDF, primary hospital, and backups. Failures set a transmission error while the active emergency UI can remain visible.

The backend validates finite numeric latitude/longitude and their ranges. It resolves hospitals, then finds an active incident within 150 m using a MongoDB geospatial query. Active means REPORTED, AMBULANCE_DISPATCHED, or ICU_RESERVED. The dedup query has no age cutoff, unlike the hospital feed.

A new incident stores the caller as primary reporter, source CITIZEN_SOS, initial condition CRITICAL_UNCONSCIOUS, hospital assignment, and a certificate record. A nearby incident is reused; another human caller is appended once as a secondary reporter. Repeating SOS as an existing reporter reuses that reporter's stored certificate digest. The create-or-merge operation is not a database transaction or atomic spatial uniqueness guarantee.

For a new incident, activateEmergencyResponse asynchronously triggers hospital notification, citizen alerts, and a corridor. PDF generation happens in the request path after persistence. A newly created incident returns HTTP 201; a merged incident returns 200. Response fields are status, incidentId, incidentCode, role, message, reporterCount, hash, timestamp, coordinates, pdfBase64, nearestHospital, and backupHospitals.

The caller polls GET /api/incidents/:id every 2.5 seconds. The client updates displayed status only inside a branch that sees ambulanceUnitAssigned; it does not unconditionally apply every returned state change. Telemetry moves in the opposite direction from VoiceTriage to the backend.

Hospital selection and capacity

Overpass searches hospital nodes/ways within 10 km, requesting up to eight elements, with a five-second query limit and four-second HTTP timeout. Named entries with coordinates are sorted by rounded Haversine distance; the first four produce one primary and up to three backups. Reported capacity is merged by hospitalId from MongoDB.

If Overpass fails or returns nothing usable, three configured Jaipur hospitals are considered within 60 km: HOSP-01 SMS Government Trauma Hospital (26.8924, 75.8150), HOSP-02 Apex (26.8530, 75.8140), and HOSP-03 Fortis Escorts (26.8480, 75.8080). Outside that coverage, the primary hospital is null. Capacity is null unless reported; it is not fabricated from map data. Selection is nearest by distance, not optimized by capacity or trauma suitability. The UI ETA formula is $\max(3, \text{round}(\text{distanceKm} * 2.2 + 2))$; it is not live traffic travel time.

Joining alerts, documents, and identity cards

Citizen notifications are broadcast to connected citizen sockets without geographic filtering. The "I'm responding" path uses the same SOS behavior rather than a separate volunteer-allocation service. There is no durable offline push-delivery system in this code.

Certificate download decodes Base64 into a PDF Blob in the browser, opens a preview, and triggers download. On native it displays a ready alert rather than implementing a complete native file-save/share workflow. If no PDF is held in memory, opening/downloading the certificate calls POST /api/sos again; requesting a document can therefore create or merge an incident.

The QR code can contain a public URL or vCard. The URL includes call identity, name, address, rounded coordinates, emergency-contact label, and blood group, but no phone-number field. The profile can still contain phone numbers locally. The card is public and query-driven, not a signed or server-verified medical identity. Its generated origin and voice socket assume port 3000.

Sources: app/App.tsx; server/services/incidents.ts; server/services/hospitals.ts; server/server.ts; server/certificate.ts.

6. Voice triage and CPR state machine

The triage engine is deterministic, not an LLM or a medical diagnosis model. TriageTree.ts defines nodes, spoken text, yes/no edges, automatic transitions, terminal actions, step labels, and three instructional image assets. VoiceTriage.tsx renders the current node, speaks it, processes voice/button answers, drives CPR pacing, and sends telemetry. The transition appendix includes every node and exact authored prompt for implementation reference only.

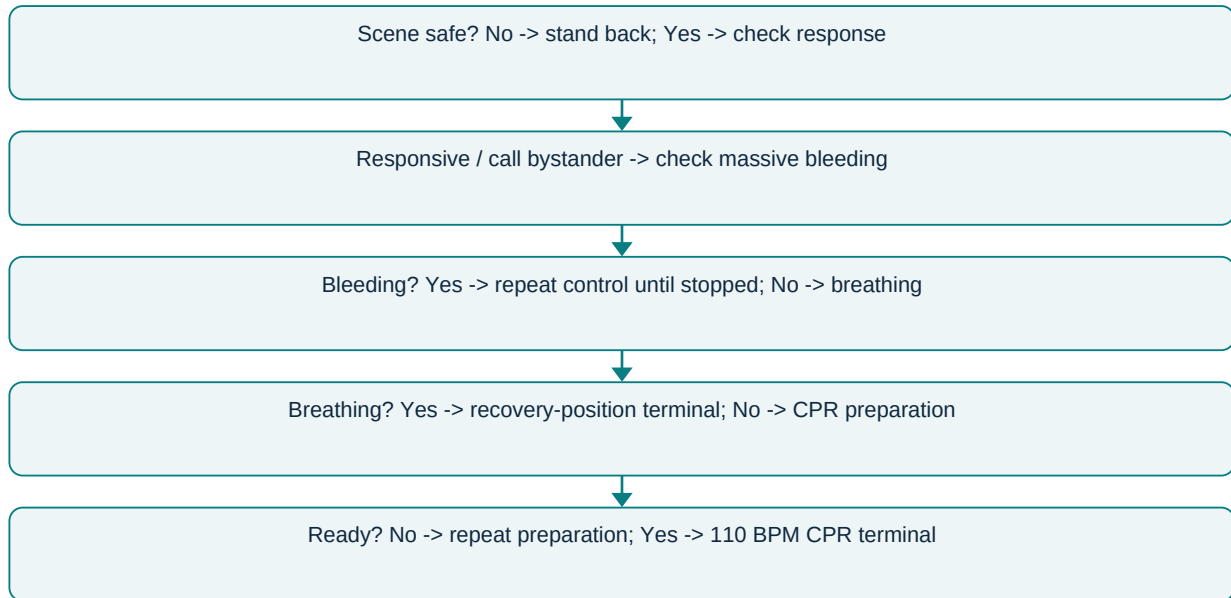


Figure 4. Application-encoded decision tree; see appendix for exact node definitions.

Scene safety branches to response assessment or a terminal stand-back instruction. Responsive and unresponsive branches converge on massive-bleeding assessment, with a bystander-call instruction on the unresponsive path. Bleeding control repeats until "yes," then moves to breathing assessment. Normal breathing goes to a recovery-position terminal node; abnormal breathing goes through CPR preparation to the CPR terminal node. These are the encoded app transitions, not a clinical endorsement.

Web input uses SpeechRecognition/webkitSpeechRecognition if available; native input uses the optional Expo recognition module. Spoken output uses browser speech synthesis and Expo Speech. The recognition locale is en-US. A repeat token allows same-node "no" answers to replay instructions and reopen listening.

CPR pacing uses 110 BPM, approximately 545 ms per interval, with a Web Audio oscillator click, animation, and native haptic feedback. The primary loop counts sets of 30 without a breath pause, sends telemetry every five beats, and sends a set-boundary update. Counters represent application-generated pacing beats, not sensor-confirmed compressions. Resume uses a smaller loop and does not reproduce all primary-loop telemetry/set behavior.

Telemetry PATCH carries victimCondition and optional cprCompressions/cprSets. Explicit mappings include unconscious response, bleeding trauma, bleeding controlled, CPR active, and recovery position. Some transitions also use node labels as condition values; the update path omits runValidators, so schema enums alone do not guarantee these updates stay within the defined condition enum.

No clinical review evidence, age-specific protocol, medical device validation, measured compression depth/rate feedback, multilingual triage tree, or emergency-service acknowledgment protocol was found. Application medical/legal claims are recorded as source behavior, not verified as current professional guidance.

7. Hospital desk flow

After server-assigned hospital login, HospitalPortal loads its hospital capacity and polls the incident feed every 2.5 seconds. It also subscribes to hospital-call and incident-created live events to refresh and display alerts. The feed contains at most 50 active incidents created in the past two hours, selected newest-first and then presented sorted by distance from the desk's supplied coordinates or the SMS fallback.

Rows include incident ID/code, primary responder ID, coordinates, distance, estimated arrival time, timestamp, victim status, dispatch status, CPR counters, assigned unit, primary reporter certificate hash, reporter count, and Google Maps directions. Server REPORTED maps to UI PENDING_DISPATCH; AMBULANCE_DISPATCHED maps to AMBULANCE_EN_ROUTE; ICU_RESERVED retains its name.

Dispatch optimistically changes the row, then PATCHes the incident with AMBULANCE_DISPATCHED and a fixed demo label, ALS Mobile Unit #108-ALPHA (Dispatched). This does not prove a real vehicle was allocated or contacted. The status route opens a corridor toward fixed SMS coordinates on dispatch and closes it on RESOLVED. Allowed status values are checked, but an ordered status-transition state machine is not enforced.

Reserve ICU checks local capacity when known, marks the incident, and changes the reported bed count through a separate request. Capacity and incident reservation are not one atomic transaction. The capacity API accepts a non-negative integer and stores the caller's bound hospital, name, count, and reporter email. Unknown capacity remains distinct from zero. Optimistic handlers do not consistently reject non-2xx HTTP responses before showing success.

The desk supports incident resolution, map links, responder voice contact, and digest inspection. The incident hash displayed is the persisted primary reporter certificate hash, not a recomputed digest. CCTV-only incidents do not mint a human certificate and may have no such hash.

Sources: app/HospitalPortal.tsx; server/server.ts; server/models/Hospital.ts; server/services/corridor.ts.

8. Control room and CCTV flow

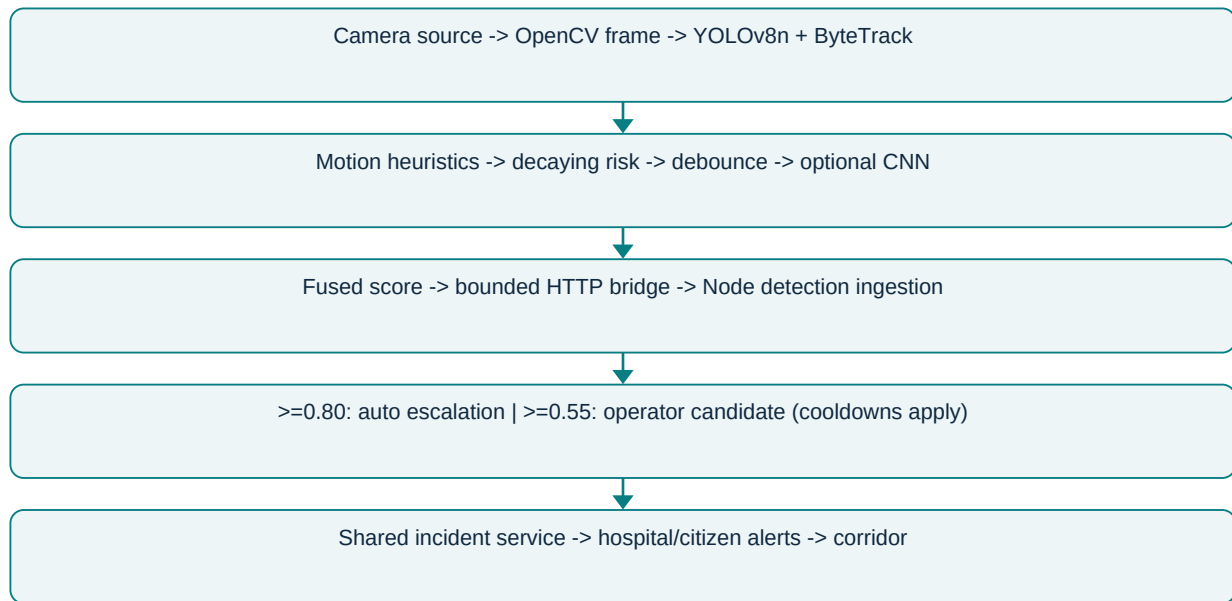


Figure 5. Detection-to-response pipeline. Confidence is not validated clinical certainty.

ControlRoom loads cameras and the most recent 40 ACCIDENT detections, refreshes REST data every 12 seconds, and consumes live detection, incident, signal, and corridor messages. Camera tiles use FastAPI MJPEG streams on web with snapshot fallback; a small clock tick refreshes fallback media. A selected event shows evidence and allows escalation. SignalSim visualizes a junction and ambulance movement from live phase input or local simulation behavior.

The Python service requests only enabled cameras from `/api/ml/cameras`. Relative filenames resolve under `ml/data/videos`; absolute file paths and RTSP/HTTP/HTTPS sources are also supported. A worker loops file footage at end-of-file. Stream read failures mark it DEGRADED and attempt reopening after one second. Missing files/open failures leave it OFFLINE. Reconnection does not explicitly restore ONLINE in the shown loop.

Defaults target 12 decoded frames/sec and inference every third frame, yielding a nominal four inference steps/sec if processing is fast enough. One YOLO tracker per camera maintains tracks and vehicle counts for cars, motorcycles, buses, and trucks, plus person detections used by collision heuristics. Actual throughput is hardware-dependent and was not benchmarked across six live cameras.

Stage 1 combines deceleration, converging box overlap, orientation change, and post-event stillness. Default weights are 0.40, 0.35, 0.10, and 0.15. Speeds are normalized by box height; vehicle/person overlap has a separate heuristic. Scene risk uses $\max(\text{current score}, \text{previous risk} * 0.96)$, preserving brief impact evidence.

Without the CNN, fused confidence is $\min(1, \text{stage1} * 0.90)$. With CNN output it is $\min(1, 0.45 * \text{stage1} + 0.55 * \text{stage2})$. Debouncing uses a six-inference window with three qualifying samples by code default. The example environment file instead sets `DEBOUNCE_REQUIRED=4`. A candidate threshold of 0.55 applies before emission, and a five-second per-pipeline guard limits repeated reports. Actual deployment settings can override these defaults.

The optional classifier converts BGR to RGB, resizes to 250 x 250, uses float32 input, and interprets class index 0 as Accident. A neighboring metadata file can verify class order; missing metadata only warns. Missing or failed CNN loading returns `stage2Score=null`. `STAGE2_THRESHOLD` is declared but not used in the inspected fusion/decision path.

Node ingestion applies X-Service-Token authentication. Confidence ≥ 0.80 can auto-escalate, with a 90-second per-camera cooldown. Confidence ≥ 0.55 can generate an operator candidate, with a separate 20-second cooldown. These timers are process-local. A high-confidence event inside auto-escalation cooldown can still enter the candidate branch. Detections are persisted before candidate broadcasts so operators receive an actionable `detectionId`.

Escalation creates/merges an incident with source CCTV_AI, synthetic reporter cctv:<cameraId>, camera ID, confidence, and snapshot. It does not mint a human responder certificate. Manual escalation rejects already-escalated events, resolves the camera, links the incident, starts the shared response path, and broadcasts in the camera zone. Dismiss currently returns success without persisting a dismissed state, so polling can show the item again.

Every ten seconds by default, camera workers enqueue heartbeats and traffic counts when a signal mapping exists. The traffic endpoint uses total count, updates the mapped approach, and sets green duration to clamp($30 + 2 * \text{vehicleCount}$, 15, 90). PREEMPTED signals are protected from this adaptive overwrite. Heartbeat fps is accepted but not stored in the Camera schema.

Sources: ml/main.py; ml/cameras.py; ml/bridge.py; ml/accident/*.py; server/routes/ml.ts; server/routes/control.ts; app/ControlRoom.tsx.

9. Green corridors and traffic simulation

A corridor can open from a new SOS, automatic/manual CCTV escalation, hospital dispatch, or a control-room request. `openCorridor` first reuses an existing ACTIVE corridor for the incident. Otherwise it requests OSRM driving geometry and duration. When routing is off or fails, it uses a two-point straight line, a 35 km/h duration estimate, and `degraded=true`.

The service projects all configured signals onto the route, keeps those within 80 m by default, orders them by distance along the polyline, and derives incoming approach from bearing. Each signal gets an ETA from route fraction with a minimum five seconds. Baseline ETA is at least one minute; optimized ETA subtracts half the adaptive-green duration per affected signal and is also floored at one minute. This saving is a formula, not measured travel-time improvement.

The corridor stores GeoJSON LineString, distances, baseline/optimized estimates, ordered signal plan, `openedAt`, and degraded state. Default lead time is 25 seconds and hold-after time is 10 seconds. The implementation immediately writes affected signals as PREEMPTED/GREEN during corridor creation, even when `holdFrom` lies in the future; a later one-second ticker tracks scheduled preemption/clearance. Thus the source is not a complete just-in-time physical traffic-control scheduler.

Closing marks CLEARED or EXPIRED, records `clearedAt`, restores ADAPTIVE mode, and removes preemption records. Expiry occurs after twice $\max(2, \text{baselineEtaMinutes})$ minutes. The ticker is started when opening a corridor, not proactively restored from persisted ACTIVE records at process boot. Overlapping corridors can overwrite a signal's single preemption slot; restoration is not coordinated across concurrent corridors.

The status route uses SMS coordinates as dispatch destination; the shared response service prefers the selected hospital and falls back to SMS when none is resolved. Existing active-corridor reuse can retain an earlier route. SignalSim is an SVG demonstration, not an adapter to roadside hardware. No vehicle GPS stream proves ambulance progress; route progress is time-based.

Sources: `server/services/corridor.ts`; `server/services/geo.ts`; `server/routes/traffic.ts`; `app/SignalSim.tsx`; `server/models/GreenCorridor.ts`.

10. Realtime, calling, and notifications

Live event channel: /ws/live

A client sends {type: subscribe, token}; the server verifies the token or uses configured development parsing, looks up the User, and records UID/role/zone. It responds subscribe-success. Event frames contain topic and payload. The frontend reconnects with exponential delay from one second capped at 30 seconds, and closes subscriptions on cleanup.

Declared topics: detection-candidate, incident-created, incident-updated, camera-status, signal-state, corridor-opened, corridor-progress, corridor-cleared, citizen-alert, and hospital-call. Not every declared topic has a current producer: normal triage/status paths primarily depend on polling rather than consistently broadcasting incident-updated, and heartbeat storage does not emit camera-status in the inspected implementation.

Zone-specific detection broadcasting targets matching control-room accounts. Role broadcasting targets hospital or citizen sockets. General broadcast excludes non-control-room users only for detection-candidate and otherwise fans out widely. Publicly exposing this channel therefore requires careful consideration of which event payloads contain locations. Subscriptions, roles, and queues have no distributed broker or replay log.

Browser voice channel: /ws

VoIP signaling is separate from authenticated live events. The server accepts register with a supplied userId and stores a process-local user/socket map. It forwards call-initiate, WebRTC offer/answer, ICE candidates, decline, hangup, and ping/pong messages. Missing/offline targets receive target-unavailable. call-accept is declared in the type union but not implemented as a switch handler.

The browser captures microphone audio, exchanges SDP/ICE through Express, and carries media peer-to-peer through WebRTC. Google STUN servers are configured; no TURN relay is provided, so restrictive network traversal is not assured. UI states are idle, incoming, initiating, ringing, connected, ended, and failed; controls include answer, decline, mute, hangup, ringtone, and duration.

Registration on /ws is not authenticated and caller IDs are client-supplied. WebRTC's encrypted media transport does not validate application identities. The public card also creates an anonymous caller ID. Both calling paths assume the current browser hostname at port 3000, rather than consistently using API_BASE. Both WebSocketServer instances attach to the same HTTP server using path options; full combined-server upgrade behavior was not exercised here.

Hospital notification and citizen alerts

The shared response function notifies hospital desks, then citizens, then opens a corridor, catching failures so they do not roll back the incident. Without all three Twilio settings, the console provider is used; partial Twilio configuration causes startup failure. A configured provider places a call using PUBLIC_BASE_URL/api/voice/twiml and attempts SMS.

DEMO_HOSPITAL_PHONE can override the registry phone. The publicly reachable TwiML route returns generated spoken XML.

Hospital-call is broadcast to all connected hospital-role sockets rather than only the assigned hospital. Citizen-alert is broadcast to every connected citizen-role socket. A query counts up to 200 citizen accounts for logging but is not an offline delivery mechanism. The payload says an ambulance is on the way even though no actual dispatch acknowledgment is required first.

Sources: server/realtime.ts; server/signaling.ts; app/liveSocket.ts; app/voipService.ts; server/services/notify.ts; server/services/response.ts.

11. Incident records, data relationships, and retention

Incident is the central aggregate: it links reporter UID strings, a hospital ID string, an optional camera ID, optional AI evidence, and one certificate entry per human reporter. DetectionEvent links cameraId and optional incidentId. GreenCorridor links incidentId and ordered signalId references. Camera can link a signal. Staff allowlists link emails to hospital or zone; User carries the effective provisioned role.

Most links are strings rather than populated Mongoose ObjectId relationships. MongoDB does not enforce relational foreign keys. IDs such as HOSP-01, OSM-<id>, JAI-CAM-014, and JAI-SIG-001 belong to separate namespaces. Human incident display codes use CAD- plus the last four ObjectId characters and are not globally unique identifiers.

The certificate digest is SHA-256(reporterId + '|' + lat + '|' + lng + '|' + server ISO timestamp). The stored certificate contains reporterId, hash, and issuedAt. It does not hash the entire mutable incident, PDF bytes, subsequent triage, or dispatch history. /api/verify/:hash validates digest syntax and looks up a stored hash; it does not recompute all incident fields or verify a cryptographic digital signature.

The incident PDF contains responder identity, coordinates, issue timestamp, hash, routed-hospital wording, a verification URL, and legal-summary content. It is not government-issued or digitally signed. The verify response discloses incidentCode, issue/record timestamps, and routedHospital but omits coordinates and responder identity.

All nine models use createdAt/updatedAt timestamps. Relevant indexes include 2dsphere locations, incident status/location, certificate hashes, unique user Firebase UID, role/login time, staff email, camera/signal/hospital IDs, detection camera/time, and corridor incidentId. Runtime schema inspection emits a duplicate certificates.hash index warning. The schema appendix includes each field, defaults, and exact declared indexes.

No TTL retention policy, archival job, immutable audit log, backup policy, incident history endpoint, or database migration framework was found. Embedded Base64 snapshots can enlarge documents; repeated detection events can grow storage. The two-hour hospital feed window filters display, not database retention. The phone field in Incident has a demo default even though the active SOS request does not collect a phone number.

12. Configuration, startup, and operations

Local launcher

run.command is a macOS-oriented Bash launcher with normal, verify, and stop modes. It checks Node/npm, discovers or starts MongoDB locally or via Docker/Homebrew, prepares environment files, locates Firebase credentials, checks project-ID consistency, installs dependencies, optionally creates the Python 3.11 environment, downloads a sample clip if missing, runs typechecks/tests, starts Express and optionally FastAPI, provisions demo roles/city infrastructure, runs the end-to-end check, and opens Expo for normal mode. Inspect it before use: it can update configuration, seed data, and reclaim ports from recognized prior processes.

Local defaults are API 3000, ML 8000, MongoDB 27017, and Expo commonly 8081. MongoDB managed by the launcher uses `~/golden-hour/mongodb`, a PID file, and a log. Process logs are `.logs/server.log` and `.logs/ml.log`. The launcher's exit trap stops the Express/ML child processes it started; its stop mode handles its MongoDB process/container. There is no deployment manager configuration in the repository.

Do not infer service readiness from a log banner alone. `/api/health` reports Express status, uptime, and auth mode, but does not actively prove current MongoDB availability. `FastAPI /health` reports worker count, online count, classifier availability, and model path; zero cameras can coexist with `status=ok`.

Manual development commands

Backend: `cd server; npm ci; configure environment; npm run dev`. App: `cd app; npm ci; configure environment; npm start` or `npm run web`. ML: create a Python 3.11 virtual environment, install `requirements.txt`, configure its matching token, then `uvicorn main:app --port 8000`. Build the server with `npm run build`; `npm start` also compiles. These commands describe supported workflows; this report did not run the launcher or modify live configuration.

Hospital provisioning uses `scripts/seed-hospital-staff.ts` with `email`, `hospitalId`, and `hospitalName`. City provisioning uses `npm run seed:city -- <staff_email>`, optionally `--if-empty`. `ensure-demo-logins.ts` and `demoLogins.ts` create/update demo Firebase staff identities when credentials exist, alongside allowlist provisioning. The default demo staff password is present in source; it is not a production access policy.

Deployment constraints

Use matching Firebase projects on client/admin sides. Expo public settings are bundled at build time and must never contain secrets. Physical phones need reachable LAN/deployed API and ML addresses. Real browser microphone/geolocation access generally requires a suitable secure origin. Configure `CORS_ORIGINS`, public verification/TwiML base URL, and both HTTP and WebSocket proxy paths. Protect ML media endpoints if footage is sensitive: their current FastAPI routes have no auth dependency and allow wildcard GET CORS.

Single-process in-memory sockets/cooldowns/timers constrain horizontal scaling. Persistent notification delivery, leader election, shared signaling, signal-conflict arbitration, and restart recovery would require additional implementation. No container image, production reverse-proxy configuration, CI deployment step, rate limiting middleware, or request-level audit system is supplied.

Configuration discrepancies

The app example says Firebase is required, but `config.ts` actually falls back to local mode when incomplete. Some README/test instructions say tests need no database, yet several current suites connect to MongoDB. The ML example sets four debounce hits while code defaults to three. `pipeline.py` contains a stale 0.85-cap comment while the actual formula uses configurable 0.90 scaling. The implementation spec names training/map/component files that are absent. This report follows executable code where comments or plans disagree.

13. Verification and known limits

Check	Observed result
App npm run typecheck	Exit 0; tsc --noEmit completed.
Server npm run typecheck	Exit 0; tsc --noEmit completed.
Mongoose model extraction	9 schemas loaded; duplicate certificates.hash index warning.
Server npm test	Exit 1: tests 37; suites 13; pass 26; fail 1; cancelled 10; skipped 0. Database-dependent checks could not connect.
Test failure evidence	MongooseServerSelectionError: connect ECONNREFUSED 127.0.0.1:27017. Initial sandbox attempt also hit listen EPERM; port-enabled retry produced the totals above.
ML sample calibration	Exit 0; 176 inferences over 21.2s; peak stage1 0.637 at 7.08s; peak fused 0.573; 1 event emitted, best fused 0.568; 0 auto-escalations. CNN absent.
End-to-end live deployment	Not run. Requires a running, correctly provisioned backend/MongoDB and can trigger notifications.

Verification scope is deliberately separate from implementation description. Typechecking proves the supplied TypeScript projects compile under their configured checks; it does not prove emergency handling, device compatibility, clinical validity, or production security. Model extraction proves the nine Mongoose schemas can load. The test and detector outputs below are recorded from this documentation task.

No live emergency was submitted, no real hospital was called, no SMS was sent, no production deployment was performed, and no live database export was read. Browser multi-role end-to-end operation, real Firebase login, real Twilio delivery, physical devices, real CCTV streams, and roadside signal hardware were not validated in this documentation task.

Material implementation limits

Location can fall back to a stale/demo coordinate. The SOS announcement claims transmission before success. Categories do not select different dispatch services. Hospital selection is proximity-based and reported capacity does not influence ranking. Dispatch is a fixed unit label, not a fleet integration. Citizen status application is conditional on a unit being assigned.

Hospital feeds/status writes are broad within the hospital role; control-room REST reads are broad across zones. VoIP registration is unauthenticated. The public civilian card interpolates query values into HTML/JavaScript and TwiML interpolates hospital text into XML without comprehensive escaping. QR data exposes profile/location in a public URL. These are source-level observations, not exploit tests or a full security audit.

The general JSON parser precedes the ML router's 2 MB parser, so a large JSON snapshot can meet Express's smaller default limit first. Detection scores and some update payloads lack comprehensive range validation. Spatial merge, bed reservation/capacity updates, manual escalation, and overlapping corridor changes are not transactional. Notifications are best-effort with no durable retries.

Green-corridor travel savings and ambulance movement are estimates/simulation. Corridor creation immediately preempts signals, while timing fields imply scheduled behavior. The ticker is not automatically restarted for existing active corridors at boot. Close broadcasts RED while database phase restoration principally resets mode/preemption. Signal conflict arbitration is absent.

The optional CNN training path is missing. The six seeded cameras reuse a single clip at different city positions; this is demonstration footage, not six independently observed real-world crashes. A single clip smoke test cannot establish precision, recall, sensitivity, false-alarm rate, or clinical usefulness.

Appendix A. Complete HTTP endpoint catalog

All explicit application routes found in Express and FastAPI source. JSON routes conventionally return status=success/error. Middleware failures generally use 401 (token) or 403 (role). Database/internal failures commonly return 500. FastAPI also exposes its standard OpenAPI/documentation routes unless disabled; no override disables them here.

GET /api/health

Access: Public. No input. status, uptime, authMode.

GET /api/voice/twiml

Access: Public. Query code, lat, lng, hospital. Returns spoken XML for Twilio.

GET /api/verify/:hash

Access: Public. 64 hex characters; success metadata, 400 malformed, 404 absent.

GET /civilian-id; /api/civilian-card

Access: Public. Query callId or id, name, address, lat, lng, blood. Returns HTML/JS identity/call card.

POST /api/auth/sync

Access: Bearer. Body displayName, optional lat/lng. Upserts token identity and returns user profile. 503 when database unavailable.

POST /api/sos

Access: Bearer. Body lat/lng. 201 new or 200 merged. Returns record, PDF, routing and reporter data.

GET /api/incidents/:id

Access: Incident reporter. No body. Own incident status, hospital/unit, own hash, timestamps.

PATCH /api/incidents/:id/triage

Access: Incident reporter. victimCondition, cprCompressions, cprSets. Returns updated incident.

GET /api/hospital/incidents

Access: Hospital. Optional lat/lng query. Active records from last 2h, max 50, distance-sorted response.

GET /api/hospitals/me

Access: Hospital. Bound hospital ID/name, capacity or null, updatedAt or null. 409 if unbound.

PATCH /api/hospitals/me/capacity

Access: Hospital. icuBedsAvailable non-negative integer. Upserts own hospital capacity.

PATCH /api/incidents/:id/status

Access: Hospital. Optional status, ambulanceUnitAssigned, icuBedReserved. Updates any incident accessible by role; dispatch/resolve corridor side effects.

GET /api/ml/cameras

Access: X-Service-Token. Enabled camera roster: ID, name, source, location, zone, approach, signalId.

POST /api/ml/detection

Access: X-Service-Token. cameraId, stage1Score, optional stage2Score, fusedConfidence, snapshotBase64, detectedAt. Returns detectionId, escalated, incidentId.

POST /api/ml/heartbeat

Access: X-Service-Token. cameraId, status, fps. Updates status and lastFrameAt; fps not persisted.

POST /api/ml/traffic

Access: X-Service-Token. cameraId, total; worker also sends counts and observedAt. Updates adaptive timing and responds with calculated data.

GET /api/control/cameras

Access: Control room. All camera metadata/status; not zone-filtered.

GET /api/control/detections

Access: Control room. Latest 40 ACCIDENT events with scores, link, snapshot, detectedAt.

POST /api/control/detections/:id/escalate

Access: Control room. No body required. Creates/merges incident and returns incidentId; 400 if already escalated.

POST /api/control/detections/:id/dismiss

Access: Control room. Acknowledges existing event; no persisted dismissal field.

GET /api/traffic/signals

Access: Control room. All signal phases, counts, timing, preemption, location.

GET /api/traffic/corridors/active

Access: Control room. ACTIVE corridors sorted by openedAt descending.

POST /api/traffic/corridor

Access: Control room. incidentId, startLat/startLng, endLat/endLng; optional zone/originHospitalId. Returns corridorId, degraded, distance/ETAs, affected signal plan.

POST /api/traffic/corridor/:incidentId/close

Access: Control room. Closes active corridor for incident; success acknowledgment.

GET ML /health

Access: Public. status, cameras, online, stage2, stage2Path.

GET ML /cameras

Access: Public. Worker snapshots: metadata, source, status, fps, stage1/fused, vehicle counts, lastError.

GET ML /snapshot/{camera_id}.jpg

Access: Public. JPEG + no-store; 404 unknown camera, 503 no frame.

GET ML /stream/{camera_id}

Access: Public. Multipart MJPEG + no-store; 404 unknown camera.

Appendix B. Complete database dictionary

Camera / cameras

Field	Type	Required / constraints / default
cameraId	String	required; unique; index=True
name	String	required
zone	String	required
location.type	String	required; enum: Point; default="Point"
location.coordinates	Array	required; default="[function]"
source	String	required
approach	String	optional / no explicit default
signalId	String	optional / no explicit default
status	String	enum: ONLINE, OFFLINE, DEGRADED; default="OFFLINE"
lastFrameAt	Date	optional / no explicit default
enabled	Boolean	default=true
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [{"cameraId": 1}, {"unique": true, "background": true}], [{"location": "2dsphere"}, {"background": true}]

Source type shape (nested members included; schema constraints above are authoritative):

```
export interface ICamera extends Document {
  cameraId: string;
  name: string;
  zone: string;
  location: {
    type: 'Point';
    coordinates: [number, number];
  };
  source: string;
  approach?: string;
  signalId?: string;
  status: 'ONLINE' | 'OFFLINE' | 'DEGRADED';
  lastFrameAt?: Date;
  enabled: boolean;
  createdAt: Date;
  updatedAt: Date;
}
```

ControlRoomStaff / control_room_staff

Field	Type	Required / constraints / default
email	String	required; unique; index=True
zone	String	required
invitedBy	String	optional / no explicit default
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [{"email": 1}, {"unique": true, "background": true}]

Source type shape (nested members included; schema constraints above are authoritative):

```
export interface IControlRoomStaff extends Document {
  email: string;
  zone: string;
  invitedBy?: string;
  createdAt: Date;
  updatedAt: Date;
}
```

DetectionEvent / detection_events

Field	Type	Required / constraints / default
cameraId	String	required; index=True
kind	String	required; enum: ACCIDENT, TRAFFIC_COUNT
stage1Score	Number	required
stage2Score	Number	optional / no explicit default
fusedConfidence	Number	required
escalated	Boolean	required; default=false
incidentId	String	optional / no explicit default
snapshotBase64	String	optional / no explicit default
detectedAt	Date	required; index=-1
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [{"cameraId": 1}, {"background": true}], [{"detectedAt": -1}, {"background": true}]

Source type shape (nested members included; schema constraints above are authoritative):

```
export interface IDetectionEvent extends Document {
  cameraId: string;
  kind: 'ACCIDENT' | 'TRAFFIC_COUNT';
  stage1Score: number;
  stage2Score?: number;
  fusedConfidence: number;
  escalated: boolean;
  incidentId?: string;
  snapshotBase64?: string;
  detectedAt: Date;
  createdAt: Date;
  updatedAt: Date;
}
```

GreenCorridor / green_corridors

Field	Type	Required / constraints / default
incidentId	String	required; index=True
status	String	required; enum: ACTIVE, CLEARED, EXPIRED
originHospitalId	String	optional / no explicit default
routeGeometry.type	String	required; enum: LineString; default="LineString"
routeGeometry.coordinates	Array	required; default="[function]"
totalDistanceKm	Number	required
baselineEtaMinutes	Number	required
optimisedEtaMinutes	Number	required
signals	Array	default="[function]"
openedAt	Date	required
clearedAt	Date	optional / no explicit default
degraded	Boolean	default=false
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [{"incidentId": 1}, {"background": true}]

Source type shape (nested members included; schema constraints above are authoritative):

```
export interface IGreenCorridor extends Document {
  incidentId: string;
  status: 'ACTIVE' | 'CLEARED' | 'EXPIRED';
  originHospitalId?: string;
  routeGeometry: {
    type: 'LineString';
    coordinates: [number, number][];
  };
  totalDistanceKm: number;
  baselineEtaMinutes: number;
  optimisedEtaMinutes: number;
  signals: {
    signalId: string;
    sequenceIndex: number;
    distanceAlongRouteKm: number;
    etaSeconds: number;
    approach: string;
    preemptedAt?: Date;
    clearedAt?: Date;
  }[];
  openedAt: Date;
  clearedAt?: Date;
  degraded?: boolean;
  createdAt: Date;
  updatedAt: Date;
}
```

Hospital / hospitals

Field	Type	Required / constraints / default
hospitalId	String	required; unique; index=True
name	String	required
icuBedsAvailable	Number	required; min=[0, 'Bed count cannot be negative']; default=0
reportedBy	String	optional / no explicit default
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [{"hospitalId": 1}, {"unique": true, "background": true}]

Source type shape (nested members included; schema constraints above are authoritative):

```
export interface IHospital extends Document {
  hospitalId: string;
  name: string;
  icuBedsAvailable: number;
  reportedBy?: string;
  createdAt: Date;
  updatedAt: Date;
}
```

HospitalStaff / hospital_staff

Field	Type	Required / constraints / default
email	String	required; unique; index=True
hospitalId	String	required; index=True
hospitalName	String	required
invitedBy	String	optional / no explicit default
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [{"email": 1}, {"unique": true, "background": true}], [{"hospitalId": 1}, {"background": true}]

Source type shape (nested members included; schema constraints above are authoritative):

```
export interface IHospitalStaff extends Document {
  email: string;
  hospitalId: string;
  hospitalName: string;
  invitedBy?: string;
  createdAt: Date;
  updatedAt: Date;
}
```

Incident / incidents

Field	Type	Required / constraints / default
location.type	String	required; enum: Point; default="Point"
location.coordinates	Array	required; default="[function]"
status	String	required; enum: REPORTED, AMBULANCE_DISPATCHED, ICU_RESERVED, RESOLVED; index=True; default="REPORTED"
primaryReporterId	String	required
secondaryReporters	Array	default="[function]"
victimCondition	String	enum: CRITICAL_UNCONSCIOUS, CPR_ACTIVE, AIRWAY_OBSTRUCTED, RECOVERY_POSITION, BLEEDING_TRAUMA, BLEEDING_CONTROLLED; default="CRITICAL_UNCONSCIOUS"
cprCompressions	Number	default=0
cprSets	Number	default=1
ambulanceUnitAssigned	String	optional / no explicit default
icuBedReserved	Boolean	default=false
assignedHospitalId	String	optional / no explicit default
assignedHospitalName	String	optional / no explicit default
phone	String	default="+91-98765-43210"
source	String	enum: CITIZEN_SOS, CCTV_AI; default="CITIZEN_SOS"
detectedByCameraId	String	optional / no explicit default
aiConfidence	Number	optional / no explicit default
snapshotBase64	String	optional / no explicit default
certificates	Array	default="[function]"
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [{"status": 1}, {"background": true}], [{"certificates.hash": 1}, {"background": true}], [{"location": "2dsphere"}, {"background": true}], [{"status": 1, "location": "2dsphere"}, {"background": true}], [{"certificates.hash": 1}, {"background": true}]

Source type shape (nested members included; schema constraints above are authoritative):

```
export type IncidentStatus = 'REPORTED' | 'AMBULANCE_DISPATCHED' | 'ICU_RESERVED' | 'RESOLVED';
export type VictimCondition = 'CRITICAL_UNCONSCIOUS' | 'CPR_ACTIVE' | 'AIRWAY_OBSTRUCTED' |
'RECOVERY_POSITION' | 'BLEEDING_TRAUMA' | 'BLEEDING_CONTROLLED';
export type IncidentSource = 'CITIZEN_SOS' | 'CCTV_AI';
```

```
export interface ICertificateRecord {
  reporterId: string;
  hash: string;
  issuedAt: Date;
}

export interface IIncident extends Document {
  location: {
    type: 'Point';
    coordinates: [number, number];
  };
  status: IncidentStatus;
  primaryReporterId: string;
  secondaryReporters: string[];
  victimCondition: VictimCondition;
```

```
cprCompressions: number;
cprSets: number;
ambulanceUnitAssigned?: string;
icuBedReserved?: boolean;
assignedHospitalId?: string;
assignedHospitalName?: string;
phone?: string;
source: IncidentSource;
detectedByCameraId?: string;
aiConfidence?: number;
snapshotBase64?: string;

certificates: ICertificateRecord[];
createdAt: Date;
updatedAt: Date;
}
```

TrafficSignal / traffic_signals

Field	Type	Required / constraints / default
signalId	String	required; unique; index=True
name	String	required
location.type	String	required; enum: Point; default="Point"
location.coordinates	Array	required; default="[function]"
approaches	Array	required; default="[function]"
currentPhase.approach	String	required
currentPhase.state	String	required; enum: GREEN, YELLOW, RED
currentPhase.endsAt	Date	required
adaptiveGreenSeconds	Number	default=30
lastVehicleCounts.N	Number	default=0
lastVehicleCounts.S	Number	default=0
lastVehicleCounts.E	Number	default=0
lastVehicleCounts.W	Number	default=0
preemption.corridorId	String	optional / no explicit default
preemption.approach	String	optional / no explicit default
preemption.holdFrom	Date	optional / no explicit default
preemption.holdUntil	Date	optional / no explicit default
mode	String	enum: ADAPTIVE, PREEMPTED, MANUAL; default="ADAPTIVE"
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [{"signalId": 1}, {"unique": true, "background": true}], [{"location": "2dsphere"}, {"background": true}]

Source type shape (nested members included; schema constraints above are authoritative):

```
export interface ITrafficSignal extends Document {
  signalId: string;
  name: string;
  location: {
    type: 'Point';
    coordinates: [number, number];
  };
  approaches: string[];
}
```

```
currentPhase: {
  approach: string;
  state: 'GREEN' | 'YELLOW' | 'RED';
  endsAt: Date;
};
adaptiveGreenSeconds: number;
lastVehicleCounts: {
  N: number;
  S: number;
  E: number;
  W: number;
};
preemption?: {
  corridorId: string;
  approach: string;
  holdFrom: Date;
  holdUntil: Date;
};
mode: 'ADAPTIVE' | 'PREEMPTED' | 'MANUAL';
createdAt: Date;
updatedAt: Date;
}
```

User / users

Field	Type	Required / constraints / default
firebaseUid	String	required; unique; index=True
email	String	required; index=True
displayName	String	required; default="Samaritan User"
role	String	required; enum: citizen, hospital, control_room; index=True; default="citizen"
hospitalId	String	optional / no explicit default
hospitalName	String	optional / no explicit default
zone	String	optional / no explicit default
lastKnownLat	Number	optional / no explicit default
lastKnownLng	Number	optional / no explicit default
lastLocation.type	String	enum: Point
lastLocation.coordinates	Array	optional / no explicit default
lastLoginAt	Date	default="[function]"
_id	ObjectId	default="[function]"
createdAt	Date	optional / no explicit default
updatedAt	Date	optional / no explicit default
__v	Number	optional / no explicit default

Declared indexes: [[[{"firebaseUid": 1}, {"unique": true, "background": true}], [{"email": 1}, {"background": true}], [{"role": 1}, {"background": true}], [{"role": 1, "lastLoginAt": -1}, {"background": true}], [{"lastLocation": "2dsphere"}, {"sparse": true, "background": true}]]]

Source type shape (nested members included; schema constraints above are authoritative):

Nested location shape (other fields are listed above):

```
lastLocation?: {
  type: 'Point';
  coordinates: [number, number];
};
```

Appendix C. Dependencies, versions, and scripts

app package.json

Package samaritan-shield 1.0.0. Entry: expo/AppEntry.js.

Dependency	Declared constraint	Lockfile resolved	Group
@expo/metro-runtime	~6.1.2	6.1.2	dependencies
@types/qrcode	^1.5.6	1.5.6	dependencies
expo	~54.0.37	54.0.37	dependencies
expo-asset	~12.0.13	12.0.13	dependencies
expo-av	~16.0.8	16.0.8	dependencies
expo-haptics	~15.0.8	15.0.8	dependencies
expo-location	~19.0.8	19.0.8	dependencies
expo-speech	~14.0.8	14.0.8	dependencies
expo-speech-recognition	^2.1.5	2.1.5	dependencies
expo-status-bar	~3.0.9	3.0.9	dependencies
firebase	^12.18.0	12.18.0	dependencies
qrcode	^1.5.4	1.5.4	dependencies
react	19.1.0	19.1.0	dependencies
react-dom	19.1.0	19.1.0	dependencies
react-native	0.81.5	0.81.5	dependencies
react-native-svg	^15.15.5	15.15.5	dependencies
react-native-web	^0.21.0	0.21.2	dependencies
@babel/core	^7.24.0	7.29.7	devDependencies
@types/react	~19.1.0	19.1.17	devDependencies
babel-preset-expo	~54.0.12	54.0.12	devDependencies
typescript	~5.9.2	5.9.3	devDependencies

Script	Command
start	expo start
android	expo start --android
ios	expo start --ios
web	expo start --web
typecheck	tsc --noEmit

Lockfile version 3; 765 non-root package-location records. These include transitive/nested packages, not necessarily unique package names. SHA-256 d34c87b23f7eebcf3091de5849e03d451e4433eefa157d97c1d572e68ec65020.

server package.json

Package samaritan-shield-server 1.0.0. Entry: dist/server.js.

Dependency		Declared constraint	Lockfile resolved	Group
cors		^2.8.5	2.8.6	dependencies
dotenv		^16.4.0	16.6.1	dependencies
express		^4.21.0	4.22.2	dependencies
firebase-admin		^14.3.0	14.3.0	dependencies
mongoose		^8.8.0	8.24.4	dependencies
pdf-lib		^1.17.1	1.17.1	dependencies
twilio		^5.10.4	5.10.4	dependencies
ws		^8.21.3	8.21.3	dependencies
@types/cors		^2.8.17	2.8.19	devDependencies
@types/express		^5.0.0	5.0.6	devDependencies
@types/node		^22.0.0	22.20.1	devDependencies
@types/supertest		^7.2.1	7.2.1	devDependencies
@types/ws		^8.18.1	8.18.1	devDependencies
supertest		^7.2.2	7.2.2	devDependencies
ts-node		^10.9.2	10.9.2	devDependencies
typescript		^5.6.0	5.9.3	devDependencies
Script		Command		
build		tsc		
start		tsc && node dist/server.js		
dev		ts-node server.ts		
test		node --require ts-node/register --test tests/**/*.test.ts		
typecheck		tsc --noEmit		
seed:city		ts-node scripts/seed-city.ts		
seed:demo-logins		ts-node scripts/ensure-demo-logins.ts		

Lockfile version 3; 380 non-root package-location records. These include transitive/nested packages, not necessarily unique package names. SHA-256 73a4f9ea0e9defb313a33918972cf7789b3d5434e2995f05abae4e2bec204ca0.

ml/requirements.txt

Requirement	Role
fastapi==0.115.6	Runtime
uvicorn[standard]==0.34.0	Runtime
opencv-python-headless==4.10.0.84	Runtime
ultralytics==8.3.55	Runtime
lap==0.5.13	Runtime
numpy==1.26.4	Runtime
httpx==0.28.1	Runtime
pydantic==2.10.4	Runtime
pydantic-settings==2.7.0	Runtime
python-dotenv==1.0.1	Runtime

ml/requirements-train.txt

Requirement	Role
tensorflow==2.18.0	Optional training
scikit-learn==1.6.0	Optional training
matplotlib==3.10.0	Optional training

Appendix D. Configuration reference

The variables below are discovered from source references and templates. Actual .env values and credentials are intentionally excluded. Python settings can be overridden by uppercase environment names; defaults are from config.py, not claims about the currently active environment.

ALLOW_INSECURE_NO_AUTH: server/middleware/auth.ts, server/middleware/serviceAuth.ts, server/tests/auth.test.ts, server/tests/corridor.test.ts, server/tests/devauth.test.ts, server/tests/traffic.test.ts

AUTO_ESCALATE_THRESHOLD: server/routes/ml.ts

CANDIDATE_COOLDOWN_SECONDS: server/routes/ml.ts

CANDIDATE_THRESHOLD: server/routes/ml.ts

CITIZEN_TOKEN: server/test_e2e_sync.js

CORS_ORIGINS: server/server.ts

DEMO_HOSPITAL_PHONE: server/services/notify.ts

DEMO_STAFF_PASSWORD: server/scripts/demoLogins.ts

ESCALATION_COOLDOWN_SECONDS: server/routes/ml.ts

EXPO_PUBLIC_ALLOW_INSECURE_NO_AUTH: app/config.ts

EXPO_PUBLIC_API_URL: app/config.ts

EXPO_PUBLIC_FIREBASE_API_KEY: app/config.ts

EXPO_PUBLIC_FIREBASE_APP_ID: app/config.ts

EXPO_PUBLIC_FIREBASE_AUTH_DOMAIN: app/config.ts

EXPO_PUBLIC_FIREBASE_PROJECT_ID: app/config.ts

EXPO_PUBLIC_FIREBASE_SENDER_ID: app/config.ts

EXPO_PUBLIC_FIREBASE_STORAGE_BUCKET: app/config.ts

EXPO_PUBLIC_ML_URL: app/config.ts

FIREBASE_SERVICE_ACCOUNT: server/middleware/auth.ts, server/scripts/demoLogins.ts, server/tests/auth.test.ts, server/tests/devauth.test.ts

GOOGLE_APPLICATION_CREDENTIALS: server/middleware/auth.ts, server/scripts/demoLogins.ts, server/tests/auth.test.ts, server/tests/devauth.test.ts

GREEN_CORRIDOR_HOLD_AFTER_SECONDS: server/services/corridor.ts

GREEN_CORRIDOR_LEAD_SECONDS: server/services/corridor.ts

GREEN_CORRIDOR_SIGNAL_RADIUS_M: server/services/corridor.ts

HOSPITAL_EMAIL: server/test_e2e_sync.js

HOSPITAL_TOKEN: server/test_e2e_sync.js

ML_SERVICE_TOKEN: server/middleware/serviceAuth.ts, server/tests/corridor.test.ts, server/tests/ml.test.ts, server/tests/traffic.test.ts

MONGO_URI: server/scripts/seed-city.ts, server/scripts/seed-hospital-staff.ts, server/server.ts, server/tests/corridor.test.ts, server/tests/incidents.test.ts, server/tests/ml.test.ts, server/tests/traffic.test.ts, server/tests/usersync.test.ts

NODE_ENV: server/middleware/auth.ts, server/middleware/serviceAuth.ts

OSRM_BASE_URL: server/services/corridor.ts, server/tests/corridor.test.ts, server/tests/ml.test.ts, server/tests/traffic.test.ts

PORT: server/config.ts, server/server.ts, server/test_e2e_sync.js

PUBLIC_BASE_URL: server/config.ts

TWILIO_ACCOUNT_SID: server/services/notify.ts

TWILIO_AUTH_TOKEN: server/services/notify.ts

TWILIO_FROM_NUMBER: server/services/notify.ts

USER: server/scripts/seed-city.ts, server/scripts/seed-hospital-staff.ts

Operational defaults and semantics

Setting	Code default	Purpose
PORT	3000	Express listener
MONGO_URI	mongodb://127.0.0.1:27017/samaritan-shield	Application database
PUBLIC_BASE_URL	http://localhost:<PORT>	Verification URL and Twilio callback origin
CORS_ORIGINS	empty list	Comma-separated browser origins; dev allows localhost ports
FIREBASE_SERVICE_ACCOUNT	unset	Raw JSON or key-file path; secret
GOOGLE_APPLICATION_CREDENTIALS	unset	Application-default credentials file path
ALLOW_INSECURE_NO_AUTH	false unless exact true	Backend dev-only unsigned bearer mode
ML_SERVICE_TOKEN	required	Shared secret checked by Node and sent by ML
AUTO_ESCALATE_THRESHOLD	0.80	Node auto-escalation score
CANDIDATE_THRESHOLD	0.55	Node and Python candidate score
ESCALATION_COOLDOWN_SECONDS	90	Per-camera Node auto escalation cooldown
CANDIDATE_COOLDOWN_SECONDS	20	Per-camera Node candidate cooldown
GREEN_CORRIDOR_SIGNAL_RADIUS_M	80	Route-to-signal distance threshold
GREEN_CORRIDOR_LEAD_SECONDS	25	Preemption lead time
GREEN_CORRIDOR_HOLD_AFTER_SECONDS	10	Hold after estimated arrival
OSRM_BASE_URL	https://router.project-osrm.org	Set off for straight-line fallback
TWILIO_ACCOUNT_SID / TWILIO_AUTH_TOKEN / TWILIO_FROM_NUMBER	unset	All three or none; partial set exits
DEMO_HOSPITAL_PHONE	unset	Optional telephony destination override
DEMO_STAFF_PASSWORD	source demo default	Provisioning password; replace outside demos
EXPO_PUBLIC_API_URL	web localhost:3000; native empty	API origin
EXPO_PUBLIC_ML_URL	web localhost:8000; native empty	Media origin
EXPO_PUBLIC_ALLOW_INSECURE_NO_AUTH	false unless exact true	Forces local client sign-in
EXPO_PUBLIC_FIREBASE_*	all six required for Firebase mode	API_KEY, AUTH_DOMAIN, PROJECT_ID, STORAGE_BUCKET, SENDER_ID, APP_ID

All Python Settings fields

Environment name	Type	Default expression
NODE_API_BASE	str	"http://localhost:3000"
ML_SERVICE_TOKEN	str	""
TARGET_FPS	float	12.0
INFER_EVERY_N	int	3
HEARTBEAT_SECONDS	float	10.0
VIDEO_DIR	Path	ML_ROOT / "data" / "videos"
YOLO_MODEL	str	"yolov8n.pt"
YOLO_CONF	float	0.35
YOLO_IMGSZ	int	640
W_DECELERATION	float	0.40
W_OVERLAP	float	0.35
W_ORIENTATION	float	0.10
W_STILLNESS	float	0.15
RISK_DECAY	float	0.96
DECEL_DROP_RATIO	float	0.65
DECEL_WINDOW_S	float	0.5
MIN_MOVING_SPEED	float	0.25
OVERLAP_IOU	float	0.15
CONVERGE_WINDOW_S	float	1.0
ORIENTATION_DELTA	float	0.40
ORIENTATION_WINDOW_S	float	1.0
STILLNESS_SECONDS	float	3.0
DEBOUNCE_WINDOW	int	6
DEBOUNCE_REQUIRED	int	3
CANDIDATE_THRESHOLD	float	0.55
STAGE1_FUSE_SCALE	float	0.90
CNN_MODEL_PATH	Path	ML_ROOT / "models" / "accident_cnn.keras"
STAGE2_THRESHOLD	float	0.70
ENABLE_STAGE2	bool	True
HOST	str	"0.0.0.0"
PORT	int	8000
JPEG_QUALITY	int	70
STREAM_FPS	float	10.0

Appendix E. Full triage node reference

Exact application-authored prompt text is reproduced below to document behavior, not as independently reviewed first-aid advice. Transitions cover every node in app/TriageTree.ts. Instructional image filenames are included where defined.

scene_safety

Step 1. Secure the scene. Look around. Are you safe to approach? No oncoming traffic, fire, or active danger to you?

onYes='check_response'; onNo='scene_not_safe'; isTerminal=false; label='SCENE SAFETY'; step=1

scene_not_safe

Stop. Do not approach. Your safety comes first. Move to a safe distance and wait for emergency services. Do not become a second victim.

action='monitor'; isTerminal=true; label='SCENE UNSAFE - STAND BACK'; step=1

check_response

Approach the victim. Shout loudly: Can you hear me? Tap their collarbone hard. Do they respond or moan?

onYes='victim_responsive'; onNo='call_108'; isTerminal=false; label='CHECK RESPONSE'; step=1

victim_responsive

They are responsive. That means they are breathing and have a pulse. Do not move their neck. Keep them completely still. Now scan for massive bleeding.

action='auto_advance'; isTerminal=false; label='VICTIM RESPONSIVE'; step=1; autoAdvanceTo='check_massive_bleeding'; autoAdvanceDelayMs=4000

check_massive_bleeding

Step 2. Check for massive bleeding now. Do a rapid visual and physical sweep of their body. Is there blood spurting like a pump, or a rapidly expanding pool of blood on the ground?

onYes='control_bleeding'; onNo='check_breathing'; isTerminal=false; label='CHECK MASSIVE BLEEDING'; step=2

control_bleeding

Massive bleeding found. Ignore everything else. Rip their clothes to expose the wound. Stuff a shirt or gauze deep into the wound cavity. Lock your elbows and put your entire body weight on it. If it is on an arm or leg, tie a tourniquet 2 inches above the wound and twist until bleeding stops completely. Has the bleeding stopped?

onYes='check_breathing'; onNo='control_bleeding_persist'; action='control_bleeding'; isTerminal=false; label='CONTROL BLEEDING'; step=2; imageSource=require('./assets/direct_pressure.jpg')

control_bleeding_persist

The bleeding has not stopped. Do NOT move to the next step. Maintain maximum pressure. Repack the wound deeper. Tighten the tourniquet further. You must stop the bleeding before anything else. Has the bleeding stopped now?

onYes='check_breathing'; onNo='control_bleeding_persist'; action='control_bleeding'; isTerminal=false; label='BLEEDING NOT CONTROLLED - KEEP PRESSURE'; step=2

check_breathing

Step 3. Check breathing now. Look at their bare chest. Is it rising and falling steadily? Put your ear next to their mouth. Can you hear air moving? Remember: occasional loud gasping means their heart has stopped. That is not normal breathing. Is the victim breathing normally?

onYes='recovery_position'; onNo='cpr_prep'; isTerminal=false; label='CHECK BREATHING'; step=3

recovery_position

They are breathing. Good. Roll them onto their side into the recovery position now. Step 1: Keep the groundward leg straight and aligned with their back. Step 2: Bend their skyward leg at the knee so the knee touches the ground for stability. Step 3: Bend their skyward arm upward and tuck their hand under their cheek to cushion the face. Step 4: Angle their face and head toward the ground so fluids drain out of the mouth, not into the airway. Monitor their breathing continuously and do not leave them on their back. Wait for paramedics.

action='recovery_position'; isTerminal=true; label='RECOVERY POSITION - MONITOR'; step=3; imageSource=require('./assets/recovery_position.jpg')

cpr_prep

They are not breathing. Their heart has stopped. You must become their physical heart. Place the heel of one hand directly in the center of their chest, between the nipples. Place your other hand on top and interlock your fingers. Lock your elbows completely straight. Are you in position?

onYes='start_cpr'; onNo='cpr_prep'; isTerminal=false; label='CPR PREPARATION'; step=4; imageSource=require('./assets/cpr_chest_compressions.jpg')

start_cpr

Starting CPR metronome now! Push down hard, at least 2 inches deep, and fast. Follow the beat. Do not stop. Let the chest recoil completely after each push. You will likely break their ribs. Ignore the cracking sound and keep pushing until paramedics physically take over or the person wakes up.

action='start_cpr'; isTerminal=true; label='CPR IN PROGRESS - COMPRESSIONS ONLY'; step=4

Appendix F. Seeded city and bundled assets

Seed definitions are demonstration infrastructure for Jaipur Central. All six cameras initially use test_video_1.mp4. These metadata positions do not establish the actual location of the bundled recording.

Traffic signals

ID / name	Coordinates [lng, lat]	Relationship / initial state
JAI-SIG-001 / Tonk Road / Gandhi Circle	[75.8055, 26.8833]	approaches N/S/E/W; ADAPTIVE
JAI-SIG-002 / Ajmer Road / 200 Ft Bypass	[75.7483, 26.8922]	approaches N/S/E/W; ADAPTIVE
JAI-SIG-003 / JLN Marg / Trauma Centre	[75.8150, 26.8924]	approaches N/S/E/W; ADAPTIVE
JAI-SIG-004 / Sikar Road / Collectorate	[75.7925, 26.9298]	approaches N/S/E/W; ADAPTIVE

Cameras

ID / name	Coordinates [lng, lat]	Relationship / initial state
JAI-CAM-014 / Tonk Road / Gandhi Circle - North approach	[75.8055, 26.8835]	approach N; signal JAI-SIG-001; OFFLINE
JAI-CAM-015 / Tonk Road / Gandhi Circle - South approach	[75.8055, 26.8831]	approach S; signal JAI-SIG-001; OFFLINE
JAI-CAM-020 / Ajmer Road / 200 Ft Bypass - East approach	[75.7485, 26.8922]	approach E; signal JAI-SIG-002; OFFLINE
JAI-CAM-022 / JLN Marg / Trauma Centre - North approach	[75.8150, 26.8926]	approach N; signal JAI-SIG-003; OFFLINE
JAI-CAM-030 / MI Road / Panch Batti - West approach	[75.8066, 26.9152]	approach W; signal -; OFFLINE
JAI-CAM-035 / Ring Road / Sitapura - South approach	[75.8340, 26.7725]	approach S; signal -; OFFLINE

Asset inventory

Videos, weights, and image assets are inventoried rather than embedding large binary contents.

Path	Bytes	Purpose
app/assets/cpr_chest_compressions.jpg	482455	Triage illustration / app icon
app/assets/direct_pressure.jpg	516747	Triage illustration / app icon
app/assets/icon.png	6	Triage illustration / app icon
app/assets/recovery_position.jpg	403475	Triage illustration / app icon
ml/data/debug/test_video_1_t07.80.jpg	109188	Calibration debug frame
ml/data/debug/test_video_1_t07.92.jpg	110449	Calibration debug frame
ml/data/debug/test_video_1_t08.04.jpg	111352	Calibration debug frame
ml/data/debug/test_video_1_t08.16.jpg	111305	Calibration debug frame
ml/data/videos/test_video_1.mp4	1473703	Looping demo footage
ml/yolov8n.pt	6549796	Object detection weights

Appendix G. Test inventory and reproducibility

server/tests/auth.test.ts

requireAuth
rejects a request with no Authorization header
rejects a non-Bearer scheme
rejects an empty bearer value
accepts a valid token and attaches the identity
identity comes from the token, not from any client-supplied field
initAuth
reports insecure-dev only under an explicit opt-in

server/tests/certificate.test.ts

incident record PDF
produces a valid PDF
carries the digest and a verification URL
names the hospital it actually routed to
says so when no hospital could be located
makes no claim of government issuance, sealing or signature
still states the responder\

server/tests/corridor.test.ts

corridor geometry
a point on the diagonal is ~0 m from the polyline
northbound traffic enters from the south approach
openCorridor
picks signals within 80 m in increasing eta order and expires them on close

server/tests/cors.test.ts

CORS in local development
accepts whichever port Expo settled on
accepts 127.0.0.1 as well as localhost
still refuses a remote origin
CORS with verified auth
the localhost widening does NOT apply
configured origins are still allowed
a request with no Origin is unaffected

server/tests/devauth.test.ts

local dev token contract
a uid derived from an email contains no colon
the same email always yields the same uid
different emails yield different uids
the server decodes the token the app mints
a signed-out app sends no token and is refused

server/tests/incidents.test.ts

createOrMergeIncident

creates a new CITIZEN_SOS incident and mints a certificate
merges a secondary CITIZEN_SOS incident and mints a second certificate
creates an AI incident with no certificate
merges an AI incident but does not mint a certificate for it

server/tests/ml.test.ts

ML Ingestion Endpoints
rejects missing service token
auto-escalates if confidence $\geq$ AUTO_ESCALATE_THRESHOLD
requires manual review if CANDIDATE_THRESHOLD $\leq$ confidence $<$ AUTO_ESCALATE

server/tests/traffic.test.ts

Traffic Control Endpoints
POST /api/traffic/corridor creates a corridor and preempts signals

server/tests/usersync.test.ts

user auth sync upsert
creates a user on first login without a lastLocation
stores a GeoJSON point only when coordinates are provided

server/tests/voip.test.ts

WebRTC VoIP Signaling Server
clients can register and exchange WebRTC signaling without leaking phone numbers
Additional integration script: server/test_e2e_sync.js synchronizes citizen/hospital accounts, checks authentication boundaries, records SOS, patches triage, checks hospital visibility and the shared digest, and follows dispatch/verification. It requires a running server and MongoDB and creates real records in the configured local environment.
CI: .github/workflows/ci.yml has server and app jobs on push and pull_request using Ubuntu and Node 22. Server runs npm ci, typecheck, and npm test; app runs npm ci and tsc --noEmit. The workflow does not provision a MongoDB service despite database-dependent suites. No ML CI job is declared.

Observed command evidence

```
app: npm run typecheck
> tsc --noEmit
exit 0

server: npm run typecheck
> tsc --noEmit
exit 0

server: npm test
tests 37 | suites 13 | pass 26 | fail 1 | cancelled 10 | skipped 0
exit 1
MongooseServerSelectionError: connect ECONNREFUSED 127.0.0.1:27017

ml: .venv/bin/python tools/calibrate.py data/videos/test_video_1.mp4 --quiet
176 inferences over 21.2s
peak stage1 0.637 at t=7.08s (overlap=1.00, deceleration=0.72)
peak fused 0.573 (stage-1 only, CNN absent)
1 event(s) emitted, best fused 0.568, 0 would auto-escalate
exit 0
```

Appendix H. Complete project file inventory

Inventory excludes installed dependencies, environments, compiled caches, generated output, and log bodies. Source/document/config-example files include byte size and a SHA-256 prefix. Local environment and credential files are named only, with contents excluded. No Git repository exists in this downloaded directory; whether a credential file was ever committed cannot be established here.

File	Bytes / SHA-256 prefix	Role
.github/workflows/ci.yml	846 / 88d97e25423fdaa2	Dependencies / configuration
.gitignore	705 / 4b97463f6c2a906e	Dependencies / configuration
README.md	13330 / 0763b4e5d9ee5d5a	Documentation
app/.env	EXCLUDED: sensitive local configuration	Credential/config presence only
app/.env.example	1563 / ea7a028a8149aff7	Dependencies / configuration
app/App.tsx	144693 / 690e9eb3bf65e971	Source / launcher
app/AuthScreen.tsx	20587 / defecf4ddf62e071	Source / launcher
app/ControlRoom.tsx	18969 / 593d40a0378f497d	Source / launcher
app/HospitalPortal.tsx	29843 / 97704a6d7a96e4de	Source / launcher
app/SignalSim.tsx	9030 / 0a3827e87fdd4e75	Source / launcher
app/TriageTree.ts	8625 / 5b0eed7f3aa7275	Source / launcher
app/VoiceTriage.tsx	30654 / 9e4207db33faa983	Source / launcher
app/api.ts	2427 / 47a0345902bb3d9a	Source / launcher
app/app.json	1634 / df9039e8fdc0e79c	Dependencies / configuration
app/assets/cpr_chest_compressions.jpg	482455 / d542c00c50d4e44e	Bundled media / weights
app/assets/direct_pressure.jpg	516747 / 13659913fd8b3d72	Bundled media / weights
app/assets/icon.png	6 / 359854b5a05053c8	Bundled media / weights
app/assets/recovery_position.jpg	403475 / f01b24d462a51579	Bundled media / weights
app/babel.config.js	107 / 2a9be300c5e2ebad	Source / launcher
app/config.ts	3576 / 6fc9fbb9a3ca288	Source / launcher
app/firebaseConfig.ts	8499 / 89b1a9282c3a56e6	Source / launcher
app/icons.tsx	8384 / c7c0aa8a33e075fd	Source / launcher
app/liveSocket.ts	2367 / c1938404f226a541	Source / launcher
app/package-lock.json	366325 / d34c87b23f7eebcf	Dependencies / configuration
app/package.json	979 / f94647c6a5baebc2	Dependencies / configuration
app/session.ts	1172 / 6ad8f41c8f4baa9c	Source / launcher
app/theme.ts	3167 / ca61ff0f81cdeac9	Source / launcher
app/tsconfig.json	528 / dbce52d94e9cef58	Dependencies / configuration
app/voipService.ts	15119 / e543fcdaf961f76b	Source / launcher
docs/ANTIGRAVITY_IMPLEMENTATION_SPEC.md	45010 / 5b302864a2c648ca	Documentation

File	Bytes / SHA-256 prefix	Role
ml/.env	EXCLUDED: sensitive local configuration	Credential/config presence only
ml/.env.example	1025 / 6adacbc5f71052d8	Dependencies / configuration
ml/.gitignore	185 / 21a687e202ced279	Dependencies / configuration
ml/README.md	1843 / b711ef11490deb20	Documentation
ml/accident/__init__.py	0 / e3b0c44298fc1c14	Source / launcher
ml/accident/classifier.py	3794 / 6d1bff728e918b70	Source / launcher
ml/accident/heuristics.py	6292 / a8149ce5bbcc75a9	Source / launcher
ml/accident/pipeline.py	4026 / b8be985ebccb120d	Source / launcher
ml/accident/tracker.py	7843 / c3610db85d51233f	Source / launcher
ml/bridge.py	3289 / 7936f0a1fc29860c	Source / launcher
ml/cameras.py	9993 / b818880f7b636836	Source / launcher
ml/config.py	4350 / 71c9d661ad0055f2	Source / launcher
ml/data/debug/test_video_1_t07.80.jpg	109188 / 4121e94f7d180469	Bundled media / weights
ml/data/debug/test_video_1_t07.92.jpg	110449 / 6c68f5460c1c4096	Bundled media / weights
ml/data/debug/test_video_1_t08.04.jpg	111352 / 2e46577dff06467	Bundled media / weights
ml/data/debug/test_video_1_t08.16.jpg	111305 / f758a3c067a7a385	Bundled media / weights
ml/data/videos/test_video_1.mp4	1473703 / 94416bb67fb3133b	Bundled media / weights
ml/main.py	4570 / 89a9063a2a7d8c05	Source / launcher
ml/requirements-train.txt	527 / 536f2678f95817b1	Dependencies / configuration
ml/requirements.txt	1277 / 708a863d02756258	Dependencies / configuration
ml/tools/calibrate.py	7801 / 2ddd87b6e788d668	Source / launcher
ml/yolov8n.pt	6549796 / f59b3d833e2ff32e	Bundled media / weights
run.command	25352 / bc6f231dda1b0207	Source / launcher
server/.env	EXCLUDED: sensitive local configuration	Credential/config presence only
server/.env.example	2329 / 1ddc3aa9edbc2886	Dependencies / configuration
server/.gitignore	46 / d5870c4d1abfddf3	Dependencies / configuration
server/certificate.ts	8942 / 4cc83d84e2adbca1	Source / launcher
server/config.ts	177 / 7a1c7be22cea2d5c	Source / launcher
server/middleware/auth.ts	6741 / 269bc3ff71576ea1	Source / launcher
server/middleware/serviceAuth.ts	1980 / e7737ca890021956	Source / launcher
server/models/Camera.ts	2493 / bc845ed56b9794b9	Source / launcher
server/models/ControlRoomStaff.ts	1125 / ffbf0b8fb7d1d28a	Source / launcher
server/models/DetectionEvent.ts	1633 / 4c634c0383fe35ea	Source / launcher
server/models/GreenCorridor.ts	2669 / 281993f42a0713b5	Source / launcher
server/models/Hospital.ts	1370 / a448bd2c87c23961	Source / launcher

File	Bytes / SHA-256 prefix	Role
server/models/HospitalStaff.ts	1483 / feba652be53b98f0	Source / launcher
server/models/Incident.ts	5430 / 6958cb025b9ab37c	Source / launcher
server/models/TrafficSignal.ts	3248 / de879ebc2a7d1ed8	Source / launcher
server/models/User.ts	2973 / 599250cbbf5496b2	Source / launcher
server/package-lock.json	166408 / 73a4f9ea0e9defb3	Dependencies / configuration
server/package.json	982 / 7f93e8b1cff79843	Dependencies / configuration
server/realtime.ts	4478 / 8471698d771c7247	Source / launcher
server/routes/control.ts	5329 / f9b82f3acf5c3ace	Source / launcher
server/routes/ml.ts	8577 / e6fb0cb0c916e9be	Source / launcher
server/routes/traffic.ts	3579 / 1d556c7a6270f250	Source / launcher
server/scripts/demoLogins.ts	2817 / 5b71e424ee532f00	Source / launcher
server/scripts/ensure-demo-logins.ts	246 / 78a3cd77a1351d40	Source / launcher
server/scripts/seed-city.ts	6417 / a3b8f06b08b0ffe3	Source / launcher
server/scripts/seed-hospital-staff.ts	3154 / bd698ccb1cce8a47	Source / launcher
server/server.ts	45686 / 86d8d2690f6d5374	Source / launcher
server/service-account.json	EXCLUDED: sensitive local configuration	Credential/config presence only
server/services/corridor.ts	10020 / ac95959980a39772	Source / launcher
server/services/geo.ts	3355 / 02ac50f05175d097	Source / launcher
server/services/hospitals.ts	7028 / 9616c7832c6455f8	Source / launcher
server/services/incidents.ts	4926 / 4026f263e54c79c9	Source / launcher
server/services/notify.ts	7962 / 53dacc3b290585c7	Source / launcher
server/services/response.ts	2007 / 119675d18a5bbf33	Source / launcher
server/signaling.ts	5767 / a792a78aeb67f10f	Source / launcher
server/test_e2e_sync.js	5877 / 2acbcb75255089b6	Source / launcher
server/tests/auth.test.ts	3186 / 24b0d343623efb96	Source / launcher
server/tests/certificate.test.ts	4061 / ac3305fe4d8a7a69	Source / launcher
server/tests/corridor.test.ts	3257 / 50fd41d88cd42cfc	Source / launcher
server/tests/cors.test.ts	2240 / 8abe5db9b3ec6381	Source / launcher
server/tests/devauth.test.ts	3223 / 05b6f220589a9dcb	Source / launcher
server/tests/incidents.test.ts	3065 / d647125ad56baaee	Source / launcher
server/tests/ml.test.ts	3169 / 6f7a316602e4f444	Source / launcher
server/tests/traffic.test.ts	3146 / eea4b3171558ec7a	Source / launcher
server/tests/usersync.test.ts	2321 / 076aba783b0a0359	Source / launcher
server/tests/voip.test.ts	4018 / be37e713287f6e13	Source / launcher
server/tsconfig.json	491 / a41ca57729077e40	Dependencies / configuration

Appendix I. Module responsibility map

Module	Responsibility
app/App.tsx	Root role switch, citizen tabs, location, SOS, QR/profile, PDF, and call overlays.
app/AuthScreen.tsx	Login/register/Google/demo user interface and profile synchronization.
app/VoiceTriage.tsx + TriageTree.ts	Voice input/output, deterministic branching, pacing and telemetry.
app/HospitalPortal.tsx	Queue, dispatch, bed capacity/reservation, resolution, digest and call controls.
app/ControlRoom.tsx + SignalSim.tsx	CCTV evidence, operator actions, sockets, and SVG junction simulation.
app/api.ts + config.ts	URLs, auth mode selection, token attachment, JSON/error wrappers.
app/firebaseConfig.ts + session.ts	Firebase/local identity, user sync, dev UID/session lifecycle.
app/liveSocket.ts + voipService.ts	Authenticated event subscription versus browser WebRTC signaling/media.
app/theme.ts + icons.tsx + assets	Design tokens, SVG icons, first-aid imagery and app icon.
server/server.ts	Application assembly, user/SOS/verification/card/triage/hospital HTTP handlers and boot.
server/config.ts + certificate.ts	Shared dedup/public URL constants and incident-record PDF layout.
server/middleware/auth.ts + serviceAuth.ts	Firebase/dev user authentication, role gates, shared-service token.
server/routes/control.ts + ml.ts + traffic.ts	Operator decisions, ML ingestion, and corridor/signal API groups.
server/services/incidents.ts + hospitals.ts	Shared create/merge/certificate operation and nearest-hospital lookup.
server/services/response.ts + notify.ts	Best-effort response fan-out; live alerts; console/Twilio provider.
server/services/corridor.ts + geo.ts	Route fallback, projections/bearings, signal scheduling, expiry/clearance.
server/realtime.ts + signaling.ts	Role/zone event delivery and separate unverified-ID signaling relay.
server/models/*.ts	Nine schemas, defaults, validation and indexes.
server/scripts/*.ts	Hospital/control staff provisioning, city/demo identities.
server/tests/*.test.ts + test_e2e_sync.js	Unit/integration contracts and manual full-flow smoke script.
ml/main.py + config.py	FastAPI lifespan/media routes and all detector settings.
ml/cameras.py + bridge.py	Threaded capture/frame publication and bounded outbound HTTP queue.
ml/accident/tracker.py	YOLO/ByteTrack, observation history, motion metrics and class counts.
ml/accident/heuristics.py + pipeline.py	Weighted collision evidence, decay, debounce, confidence fusion, emit guard.
ml/accident/classifier.py	Optional trained CNN load/class-order verification/preprocessing/inference.
ml/tools/calibrate.py	Offline real-pipeline video analysis, event statistics, optional annotated dumps.
run.command + .github/workflows/ci.yml	Local setup/verification/launch and CI commands.
docs/ANTIGRAVITY_IMPLEMENTATION_SPEC.md	Design/build plan; not an authoritative statement of implemented features.

Appendix J. Plan versus implemented checkout

Design / documentation idea	Actual checkout
Stage 2 CNN training	Runtime optional loader exists; ml/train/build_model.py and train_cnn.py are absent; accident_cnn.keras absent.
Map and split control components	No map SDK dependency or planned control/CityMap.tsx. ControlRoom.tsx and SignalSim.tsx implement the current dashboard. Citizen MAPS is a styled coordinate panel.
Nearby geofenced citizen push	Current implementation broadcasts live alerts to every connected citizen; no 2 km targeting or durable offline push service.
Authentic dispatch and signal control	Fixed ambulance label; database traffic phases and SVG simulation. No hardware/fleet connector.
Signed/legal protection document	Persisted SHA-256 record lookup and generated PDF; no digital signature or government issuance.
All API access protected	Health, verification, public civilian card and TwiML are public; ML media is public; /ws registration is not authenticated.
No-database test suite	Current incident, corridor, traffic, ML, and user-sync tests depend on MongoDB.
Capacity-aware destination	Capacity is enriched for display; nearest-hospital selection remains distance-based.

Recommended implementation order

Before real-data deployment, correct location acknowledgment, access boundaries, public-card escaping, and signaling authentication. Then make dispatch and notifications durable and hospital-scoped, introduce transactional capacity/state updates and restart-safe corridor ownership, and validate the detector against representative labeled footage. Independently review the encoded clinical/legal content. No application code changes are included in this deliverable.

End of report. Every described implementation is tied to the supplied source snapshot. Re-run verification and regenerate this reference after changing code, schemas, dependencies, or configuration.